

Questions/Answers: Stakeholder

- 1) Reason for cube shape?
 - a) Allows for 2 heights of children
 - i) Flips so actual chair sitting part isn't in the middle of the cube but ends up raising depending on the side
- 2) Problems with current design?
 - a) Only has 2 heights
 - i) Best for attention is having feet directly flat on floor
(1) 90-90
 - b) Tray is quite large even with the folding
 - c) Cost
 - i) Expensive
- 3) Activities done in the chair?
 - a) No cushion
 - b) Used for feeding, speech, handwriting, fine motors, just for sitting
 - c) Can use handles to push yourself up
 - d) Back support
- 4) How is it used in groups or individuals?
 - a) Mostly 1 on 1 in therapy world
 - i) Can be used for "circle time"
(1) Provides barrier to prevent kids from touching each other
- 5) Design features to prevent injury?
 - a) Cube shape used to prevent it from tipping possibly
 - b) Tray makes it more stable
 - i) Fully surrounded with tray
 - c) Plastic is easily cleanable
 - i) Hard to clean fabric
- 6) How does it support people struggling with physical limitations?
 - a) With more severe impairments it doesn't help
 - b) Needs to be able to sit in the chair on their own
 - c) Therapist holds the child with impairment to help them sit and stay seated
 - d) Mainly hard because of its one size fit all
- 7) What type of children is it for?
 - a) More for children with typical abilities/Higher level physical abilities
 - b) Learning disabilities, fine motor issues, feeding issues
- 8) Does it need to be portable?
 - a) Could be either
 - i) Currently isn't portable
 - ii) Could be useful to take around
(1) Changing environments for kids is important
- 9) Average size? Different sizes for different kids?
 - a) No different sizing only 1 type
- 10) Functionality, aesthetic, and comfort what is most important?
 - a) Functionality most important so family can use every day
 - b) Cost is important

- c) Doesn't look outdated or bad in the home
 - d) Comfort isn't too important to kids matters more the activity
 - i) Rounded bottom to mold to your bottom
 - ii) Hard plastic
- 11) How does the child enter or exit?
- a) Most kids sit in it and tray is placed on top
 - b) Some kids climb up between the chair and tray
 - i) Okay cuz sides support the kid so they wont slide off
- 12) Why is it expensive?
- a) Because its labeled medical its more expensive
 - b) Tray has some folding mechanics that could make it more expensive (metal)
 - c) Special plastic mechanics to attach together
- 13) Hollow or solid plastic?
- a) Its hollow plastic
- 14) Common complaints from kids?
- a) No common complaints from kids
- 15) Needed features?
- a) Stability - Back support
 - b) Different heights
 - c) Tray
- 16) Current way of storage for the chair?
- a) Sit outside in the hallways
 - b) Could be improved
 - i) Doesn't fold
 - c) Tray is bulky and big during storage
- 17) What restricts portability?
- a) Size
 - b) Lack of foldability
 - c) Tray is big
- 18) How would you clean? Is it difficult?
- a) Use cleaning wipes
 - b) Very easy to clean
 - c) Not much wear and tear on the actual chairs
- 19) What parts on the chair have ever broken?
- a) Has only ever split along the plastic melding seam
 - i) Not common, only once
- 20) Safety concerns due to common habits of kids?
- a) Throwing themselves backward in the chair
 - i) Causing it to tilt
 - ii) Tray helps with stability a bit
 - b) Kicking into the tray
 - i) Tray is light enough not to hurt the child
- 21) Texture of the chair and its effect?
- a) Rough plastic on chair

- b) Tray is a little smoother
 - c) Doesn't effect anything
- 22) Is the tray lifting a common issue?
 - a) Possibly
 - i) Very easy to lift
 - b) Therapist might keep their hand on the tray
- 23) What age group is it for?
 - a) Great for younger kids that typical chair and table wouldn't work for
 - i) Preschool age/Early elementary
 - b) Doesn't know about weight limits?
- 24) Trays use?
 - a) Attaches to chair
 - b) Has lip to keep items from falling off
 - c) Solid locking mechanism on legs so it doesn't collapse
- 25) Cost?
 - a) Cheaper (Not \$140 just for tray)
 - b) Affordable for average person and households

Notes Based on Questions/Answers

Design Requirements & Key Takeaways

Project Specifications (Must Meet)

- **Cost:** Chair + tray $\leq$ \$100
- **Weight:** $\leq$ 8 lb
- **Seat Height:** Adjustable from 2.5-4.5 in
- **Weight Capacity:** Supports 26 lb
- **Scale:** Build a 1:2 scale prototype (design for full-scale production)
- **Cost Analysis:** Based on plastic, even if the prototype is made from wood.

Essential Design Requirements

Adjustability

- Adjustable seat height (2.5–4.5 in).
- Support proper ergonomics:
 - Feet flat on floor.
 - 90° ankles, knees, and hips (90-90-90).

Stability & Safety

- Wide, stable base to resist tipping.
- Secure tray attachment that is difficult for children to lift accidentally.
- Side support to help keep children seated safely.

Tray

- Detachable and lightweight.
- Stable work surface with a raised lip.
- Compact enough for easier storage.

Portability

- Lightweight enough for children and therapists to move.
- Compact for storage and transport.

Cleaning & Durability

- Smooth plastic surfaces for easy cleaning.
- Durable enough for daily therapy and classroom use.

Cost

- Simple, affordable design that can be manufactured for $\leq$ \$100.

Intended Users & Use

- **Users:** Preschool to early elementary children with mild physical, feeding, learning, or fine motor challenges.
- **Settings:** Home, school, and pediatric occupational therapy.
- **Activities:** Feeding, handwriting, speech therapy, fine motor tasks, puzzles, worksheets, and circle time.
- **Design Goal:** Support child independence while providing safe, ergonomic seating.

Plans

POSTER: <https://canva.link/4g8ei9f673fvyz4>

Prototype in Writing for CAD:

Chair Body

- Hollow plastic cube with rounded corners.
 - Reinforced ribs on the inside instead of thick plastic to reduce weight and add structure and stability
- Wide base for stability.
 - Possibly a back triangle piece to add tipping protection
- Open front so children can climb in easily.
- Smooth surfaces for easy cleaning.
 - Basically just plastic menial padding
 - Curved for better cushioning without fabric

Adjustable Seat (Main Innovation)

Instead of two heights from flipping the chair:

- Seat slides vertically inside the cube.
- Four or five preset positions (2.5–4.5").
- Locks into place with spring-loaded pins or push buttons on both sides.
 - Handle that you pull on both sides and then release once you reach the desired hole at the desired height
 - Back has an insert non spring loaded just slide a bar in throughout the main portion of the seat

Tray attachment

- Plastic tray with two hooks
 - Hooks located at the back of the arms of the tray which sit on top of the arms of the chair
 - Actually latches into the tray with these c-hooks so it doesnt fly up if it gets hit
- Raised lip around the edge.
- Folds out and locks legs of the tray in place with a hinging plastic piece in a diagonal placement that extends the width of the tray

Weight Reduction

Instead of solid plastic:

- Hollow shell.
- Internal rib structure (like inside a cooler or storage bin).
- Honeycomb ribs underneath the seat.

- Creates a cushiony structure in the plastic seat while remaining light and easy to clean

Storage

Instead of folding: Design chairs so they nest together/stack to take up less space, trays fold flat and lay in a designated holder

Folding: Tray is spring loaded hooks, slide it out. The sides fold inwards to make both of the tray and chair flat and then a designated holder that comes with it holds them together..

Safety

- Rounded corners everywhere.
- Wide bottom.
- Low center of gravity.
- Tray locks positively.
- Seat locks into fixed positions.

Script (AI Overview)

Slide 1 – Our Mission (Rene)

"Our mission was to design and build an affordable, lightweight cube chair with a detachable tray for use in homes, schools, and occupational therapy settings.

The purpose of the chair is to encourage children's independence while also promoting proper posture, comfort, and safety during activities like learning, eating, or playing.

To meet our design requirements, the chair needed to cost less than one hundred dollars, support up to twenty-six pounds, weigh less than eight pounds, and include an adjustable seat height. Since this is a prototype, we created it at half scale while still maintaining these key design features."

Slide 2 – Prototype 1(Avery)

"Our prototype focused on improving the original design by making it more functional and convenient.

One of its main features was a multi-height adjustable seat, allowing the chair to accommodate children of different sizes while improving comfort and ergonomics.

Unlike the previous design, which had a fixed seat height, this prototype gave users flexibility as the child grows.

We also designed the chair to fold, making it easier to store when it's not being used and improving portability for families or therapists who may need to transport it.

Another improvement was integrating the table directly with the chair by attaching it underneath the seat. This reduced the number of separate pieces and made the overall design more compact.

Overall, Prototype 1 improved the chair's functionality, versatility, and ease of use."

Slide 3 – Prototype 2 (Dylan)

"Our second prototype focused even more on portability and compact storage.

The chair includes extendable side panels that retract, allowing the chair to fold into a much smaller form.

Like Prototype 1, it also features a multi-height adjustable seat, but this time it uses sliding support rods that lock securely into different height positions.

Instead of attaching the table permanently, this design uses a removable table that can be disassembled when transporting or storing the chair.

These improvements reduce the overall size of the chair when folded, making it easier to carry and store while still maintaining structural stability.

By combining foldability and adjustability, Prototype 2 became a practical solution for classrooms, homes, and therapy environments."

Slide 4 – Prototype 3 (Alex)

"Our third prototype explored a completely different approach to the tray system.

Instead of removing the tray, it folds completely inside the chair.

To make this possible, the left side of the chair was widened and hollowed out to create a storage compartment.

A pivot arm connects the tray to the chair, and a locking-pin mechanism secures it at ninety degrees to provide a stable writing surface.

The tray itself can rotate up to two hundred seventy degrees using a secondary pivot, allowing it to fold neatly into the storage compartment.

Although this design was very compact and innovative, it required many moving parts and a much more complicated mechanism."

Slide 5 – Elimination Process & Final Design (Xavi)

"After developing all three concepts, we evaluated each design based on functionality, safety, ergonomics, portability, storage efficiency, and ease of manufacturing.

Prototype 1 provided a foldable chair with an integrated table and adjustable seat. However, attaching the table beneath the seat made the design more difficult to store and maintain.

Prototype 2 improved portability through its sliding side panels, foldable frame, and removable table. However, having a separate table meant there was a higher chance of pieces being misplaced.

Prototype 3 offered an innovative rotating tray that stored inside the chair, but the mechanism required many additional moving parts, increasing manufacturing cost, assembly time, maintenance, and the possibility of mechanical failure.

Rather than choosing one design, we decided to combine the strongest features from Prototypes 1 and 2 into a hybrid design.

Our final prototype includes a multi-height sliding seat, a foldable chair for compact storage, sliding side supports for increased stability, adjustable support holes in the backrest for better child positioning, a tilt support to improve safety, and a rear-mounted tray attachment.

We chose to eliminate Prototype 3 because its rotating tray mechanism added unnecessary complexity without providing enough additional benefits. A rear-mounted tray provides the same functionality while being simpler, more reliable, easier to manufacture, and more cost-effective.

Overall, our final hybrid design successfully meets our project goals by creating a chair that is safe, adjustable, portable, and practical for children in a variety of learning environments.

Thank you for listening. We'd be happy to answer any questions."